

Pandas vs Power Query (M)

A practical side-by-side reference for data transformation, cleaning, aggregation, joins, dates, and analytics

#	Data Operation	Pandas / Python	Power Query / M
1	Import Pandas	<code>import pandas as pd</code>	Power Query is built into Power BI
2	Read CSV	<code>pd.read_csv("sales.csv")</code>	<code>Csv.Document(File.Contents("sales.csv"))</code>
3	Create DataFrame	<code>df = pd.DataFrame(data)</code>	<code>#table(...)</code>
4	View first rows	<code>df.head()</code>	<code>Table.FirstN(Source, 5)</code>
5	View last rows	<code>df.tail()</code>	<code>Table.LastN(Source, 5)</code>
6	Number of rows	<code>df.shape[0]</code>	<code>Table.RowCount(Source)</code>
7	Number of columns	<code>df.shape[1]</code>	<code>List.Count(Table.ColumnNames(Source))</code>
8	Column names	<code>df.columns</code>	<code>Table.ColumnNames(Source)</code>
9	Data information	<code>df.info()</code>	<code>Table.Schema(Source)</code>
10	Statistics	<code>df.describe()</code>	<code>Table.Profile(Source)</code>
11	Select one column	<code>df["Sales"]</code>	<code>Table.SelectColumns(Source, {"Sales"})</code>
12	Select multiple columns	<code>df[["Name", "Sales"]]</code>	<code>Table.SelectColumns(Source, {"Name", "Sales"})</code>
13	Filter rows	<code>df[df["Sales"] > 1000]</code>	<code>Table.SelectRows(Source, each [Sales] > 1000)</code>
14	Multiple filters AND	<code>df[(df["Sales"] > 1000) & (df["Region"] == "South")]</code>	<code>Table.SelectRows(Source, each [Sales] > 1000 and [Region] = "South")</code>
15	Multiple filters OR	<code>df[(df["Region"] == "South") (df["Region"] == "North")]</code>	<code>Table.SelectRows(Source, each [Region] = "South" or [Region] = "North")</code>
16	Query/filter syntax	<code>df.query("Sales > 1000")</code>	<code>Table.SelectRows(Source, each [Sales] > 1000)</code>
17	Select by position	<code>df.iloc[0:5, 0:3]</code>	<code>Table.FirstN(Source, 5) + column selection</code>
18	Rename column	<code>df.rename(columns={"Sales": "Revenue"})</code>	<code>Table.RenameColumns(Source, {{"Sales", "Revenue"}})</code>
19	Add column	<code>df["Revenue"] = df["Qty"] * df["Price"]</code>	<code>Table.AddColumn(Source, "Revenue", each [Qty] * [Price])</code>
20	Conditional column	<code>df["Cat"] = df["Sales"].apply(lambda x: "High" if x > 1000 else "Low")</code>	<code>Table.AddColumn(Source, "Cat", each if [Sales] > 1000 then "High" else "Low")</code>
21	Multiple conditions	<code>df["Cat"] = df["Sales"].apply(classify)</code>	<code>if [Sales] >= 10000 then "High" else if [Sales] >= 5000 then "Medium" else "Low"</code>
22	Sort ascending	<code>df.sort_values("Sales")</code>	<code>Table.Sort(Source, {{"Sales", Order.Ascending}})</code>
23	Sort descending	<code>df.sort_values("Sales", ascending=False)</code>	<code>Table.Sort(Source, {{"Sales", Order.Descending}})</code>

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24	Sort multiple columns	df.sort_values(["Region","Sales"])	Table.Sort(Source, {{"Region", Order.Ascending}, {"Sales", Order.Descending}})
25	Remove duplicates	df.drop_duplicates()	Table.Distinct(Source)
26	Remove duplicates by column	df.drop_duplicates(subset=["CustomerID"])	Table.Distinct(Source, {"CustomerID"})
27	Check nulls	df.isnull().sum()	Table.Profile(Source) / null checks
28	Remove null rows	df.dropna()	Table.SelectRows(Source, each [Sales] <> null)
29	Replace null	df["Sales"].fillna(0)	Table.ReplaceValue(Source, null, 0, Replacer.ReplaceValue, {"Sales"})
30	Mean for null	df["Sales"].fillna(df["Sales"].mean())	Calculate average + Table.ReplaceValue()
31	Unique values	df["Region"].unique()	List.Distinct(Source[Region])
32	Count unique	df["Region"].nunique()	List.Count(List.Distinct(Source[Region]))
33	Frequency count	df["Region"].value_counts()	Table.Group() + Table.RowCount()
34	GroupBy + Sum	df.groupby("Region")["Sales"].sum()	Table.Group() + List.Sum()
35	GroupBy + Mean	df.groupby("Region")["Sales"].mean()	Table.Group() + List.Average()
36	GroupBy + Min	df.groupby("Region")["Sales"].min()	Table.Group() + List.Min()
37	GroupBy + Max	df.groupby("Region")["Sales"].max()	Table.Group() + List.Max()
38	GroupBy + Count	df.groupby("Region")["Sales"].count()	Table.Group() + Table.RowCount()
39	Multiple aggregations	df.groupby("Region")["Sales"].agg(["sum","mean","max"])	Table.Group() with multiple aggregations
40	Group by multiple columns	df.groupby(["Region","Product"])["Sales"].sum()	Table.Group(Source, {"Region","Product"}, ...)
41	Inner Join	pd.merge(df1, df2, on="ID")	Table.NestedJoin(..., JoinKind.Inner)
42	Left Join	pd.merge(df1, df2, on="ID", how="left")	Table.NestedJoin(..., JoinKind.LeftOuter)
43	Right Join	pd.merge(df1, df2, on="ID", how="right")	Table.NestedJoin(..., JoinKind.RightOuter)
44	Full Outer Join	pd.merge(df1, df2, on="ID", how="outer")	Table.NestedJoin(..., JoinKind.FullOuter)
45	Append rows	pd.concat([df1, df2])	Table.Combine({Table1, Table2})
46	Convert date	pd.to_datetime(df["Date"])	Table.TransformColumnTypes(..., {"Date", type datetime})

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47	Extract year	df["Year"] = df["Date"].dt.year	Date.Year([Date])
48	Extract month	df["Month"] = df["Date"].dt.month	Date.Month([Date])
49	Month name	df["MonthName"] = df["Date"].dt.month_name()	Date.MonthName([Date])
50	Day	df["Day"] = df["Date"].dt.day	Date.Day([Date])
51	Date difference	(df["End"] - df["Start"]).dt.days	Duration.Days([End] - [Start])
52	Time difference	(df["End"] - df["Start"]).dt.total_seconds()	Duration.TotalSeconds([End] - [Start])
53	Previous row	df["Previous"] = df["Sales"].shift(1)	Index + self-merge
54	Difference from previous row	df["Sales"] - df["Sales"].shift(1)	Index + self-merge + subtraction
55	Moving average	df["Sales"].rolling(7).mean()	Index + List.Range() + List.Average()
56	Pivot	pd.pivot_table(df, values="Sales", index="Region", columns="Product", aggfunc="sum")	Table.Pivot()
57	Unpivot	df.melt(...)	Table.Unpivot()
58	Transform column	df["Sales"].apply(lambda x: x * 1.1)	Table.TransformColumns()
59	String uppercase	df["Name"].str.upper()	Text.Upper([Name])
60	String lowercase	df["Name"].str.lower()	Text.Lower([Name])
61	String length	df["Name"].str.len()	Text.Length([Name])
62	Contains text	df["Name"].str.contains("John")	Text.Contains([Name], "John")
63	Replace text	df["Name"].str.replace("Old", "New")	Text.Replace([Name], "Old", "New")
64	Split column	df["Name"].str.split(" ", expand=True)	Table.SplitColumn()
65	Index column	df.reset_index()	Table.AddIndexColumn()
66	Convert to records	df.to_dict(orient="records")	Table.ToRecords(Source)
67	Export CSV	df.to_csv("output.csv")	Usually handled outside Power Query
68	Export Excel	df.to_excel("output.xlsx")	Usually handled outside Power Query